Predicting Future Forest Fires Rubric

DS 4002 – Instructor: Christina Barton
Due: tbd
Submission format: GitHub repository (submitted by link to canvas)

Individual Assignment

Why am I doing this? The goal of this case study is to allow you to explore a real life application of what you have learned through this class working with time series data. You will gain insight to how logistic regression and random forest classifiers work as well as their benefits and differences. You will also practice validating results and making predictions using the model you create.

What am I going to do? You will independently develop a machine learning system that predicts whether a wildfire will occur on a given day in California using weather and environmental time series data. You will first collect and clean time series data, ensuring that variables are properly formatted for analysis. Then, you will implement both a Logistic Regression model and a Random Forest model to classify whether a wildfire will occur. You will evaluate the performance of each model and compare their effectiveness in predicting wildfire events.

Tips for success:

- Examine the data thoroughly before beginning work and identify any patterns or balance issues (also identify if there appears to be any strange data that needs removal)
- Keep your code and documents organized for easier access to your work
- Keep in mind the evaluation metrics you are using and ensure all of your grounds are covered
- Make your results easy to understand and interpretable

How will I know I have succeeded? You will meet expectations on Predicting Future Forest Fires when you follow the criteria in the rubric below.

Spec Category	Spec Details
Formatting	- One Github Repository should contain: - A README.md file - A LICENSE.md file - A SCRIPTS folder containing any code used in the case study - A DATA folder that includes both the cleaned and original data sets used - AN OUTPUT folder that includes any visuals produced during processing and modeling - Write-Up pdf that explains more specific findings and conclusions

<table>
<tr><td>README.md</td><td>Provide a guide for an outsider trying to understand your project and should include:
Structure and where all documents are located in the repository
Software and platforms required for this study
Step by step instructions of how to recreate your results</td></tr>
<tr><td>LICENSE.md</td><td>Use a standard GitHub license (MIT is recommended)</td></tr>
<tr><td>SCRIPTS</td><td>Preprocessing script should:
Load raw data
Clean the dataset (handle missing values, remove errors)
Format time series data correctly (dates, ordering)
Create/define the target variable
Wildfire occurrence
Output a clean dataset for modeling
Modeling script should:
Load the cleaned dataset
Split data into training/testing sets
Train a Logistic Regression model
Train a Random Forest model
Evaluate both models using Accuracy, Precision, Recall, F1
Compare model performance
Generate predictions</td></tr>
<tr><td>Write up</td><td>Introduction
Background, context, and project overview
Data description
Where data came from and key variables
Challenges
Difficulties in coding or data collection
Methodology
Explain logistic regression and random forest classification and why these models are chosen
Results
Model performance and comparison between models
Visualizations
Predictions
Explain what the model predicted for the first few days of 2024 and interpret
Conclusions
Summary of findings, limitations, and improvements</td></tr>
<tr><td>References</td><td>All references should be listed at the end of the document
Use IEEE Documentation style</td></tr>
</table>

Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric. This structure is pulled from Streifer & Palmer (2020).